A Unified Approach to Detect Clouds, Water, and Shadows in Hyperspectral Satellite Images: A Focus on PRISMA Product Enhancement

Salvatore Maresca and Stefania Matteoli, *Senior Member, IEEE*

***Abstract*—The recent advancements in hyperspectral satellite missions have significantly expanded Earth observation capabilities. Availability of high-resolution hyperspectral data and products thereof, such as those delivered by the PRISMA mission of the Italian Space Agency (ASI), provides unprecedented opportunities for monitoring Earth processes in a wide range of applications. Enhancement of hyperspectral image products plays a crucial role in further increasing data exploitation possibilities. In this work, we propose a unified, physics-based approach for joint detection of clouds, water, and shadows in hyperspectral images (APACHE), focusing on PRISMA L1 product enhancement. Unlike existing approaches that typically address the detection of one individual surface type at a time, our methodology jointly extracts the three surfaces of interest in an automated, stepwise, and unsupervised manner. More importantly, there is no need for parameter tuning, labeled data, or multisource images. APACHE only leverages spectral properties of clouds, water, and shadows combined with a nonparametric clustering technique. Experimental results on a variety of PRISMA images from diverse datasets demonstrate the effectiveness of the proposed methodology in accurately detecting the surfaces of interest, both with respect to state-of-the-art methods and in enhancing PRISMA products. On average, APACHE achieves accuracies of 96.42% for cloud detection, 99.23% for water detection, and 97.55% for shadow detection, with improvements ranging from 2% to 30% over the best-performing comparison method. These results highlight APACHE's potential to improve hyperspectral satellite data exploitation across a wide range of remote sensing applications.**

***Index Terms*—Cloud detection, hyperspectral, PRISMA, shadows, water detection.**

I. Introduction

BY PROVIDING a spectrum for each pixel across hundreds of narrow, contiguous spectral bands [typically from the visible to the near and short-wave infrared (VIS-NIR-SWIR)], hyperspectral imaging has become a critical technology for identifying surface materials and analyzing atmospheric composition [1]. Its wide-ranging applications, from geology and hydrology to precision agriculture, defense, urban mapping, and climate studies, demonstrate its key role in various scientific fields [1].

While hyperspectral sensors have been traditionally deployed on airborne and ground-based platforms, recently, there has been growing interest in spaceborne hyperspectral missions [2]. These missions offer the potential to globally and continuously monitor Earth with hyperspectral capabilities, enabling more detailed environmental mapping and a deeper understanding of processes related to the environment and human activities. Several years after the pioneering spaceborne hyperspectral missions, such as CHRIS on PROBA-1 [3] and Hyperion on EO-1 [4], there has been a renewed momentum with the launches of Gaofen satellites [5], "*Precursore iperspettrale della missione applicativa*" (PRISMA) [6], and EnMAP [7], along with installation of several hyperspectral sensors aboard the international space station (ISS), such as DESIS [8] and EMIT [9].

Among these, PRISMA, developed by the Italian Space Agency (ASI), has been attracting a great deal of interest. Since its launch in 2019, PRISMA has offered users several data levels and products thereof, in a framework of global coverage and almost open-data policy, thus unlocking numerous possibilities for data exploitation across a wide range of applications [10], [11], [12], [13]. PRISMA Level 1 (L1) products provide radiometrically corrected and calibrated top-of-atmosphere (ToA) spectral radiance data, along with land-use masks, including cloud and water masks. Level 2 (L2) products offer bottom-of-atmosphere (BoA) reflectance data, which also include maps of atmospheric constituents [14].

Within this framework, designing processing chains for PRISMA products enhancement plays a crucial role in expanding its potential in data exploitation and simplifying its use across diverse applications and users. For instance, in [15], L1 products were refined by improving their georeferencing accuracy, and the work in [16] adopted similar strategies for L2 products. Preliminary results in [16] also provided experimental evidence that enhancing PRISMA L1 products—by refining cloud and water masks and incorporating a shadow mask—is

Received 7 August 2025; revised 3 November 2025; accepted 17 November 2025. Date of publication 21 November 2025; date of current version 11 December 2025. This work was supported by Consiglio Nazionale delle Ricerche through the project "AHEAD-Advancing Hyperspectral Satellite Remote Sensing Toward a Digital Twin Earth" under Project DIT.AD001.232. *(Corresponding author: Stefania Matteoli.)*

The authors are with the Consiglio Nazionale delle Ricerche (CNR), Istituto di Elettronica e di Ingegneria dell'Informazione e delle Telecomunicazioni (CNR-IEIIT), 56122 Pisa, Italy (e-mail: salvatore.maresca@cnr.it; stefania.matteoli@cnr.it).

This article has supplementary downloadable material available at https://doi.org/10.1109/TGRS.2025.3635824, provided by the authors.

Digital Object Identifier 10.1109/TGRS.2025.3635824

© 2025 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License.
For more information, see https://creativecommons.org/licenses/by/4.0/

both feasible and worthy of further investigation. Availability of such accurate masks is indeed critical for end-users, as water identification is vital for water-related applications, while masking out water, cloud, and shadow pixels—due to their peculiar spectral characteristics—is required by many hyperspectral processing tasks (e.g., for preventing contamination of second-order statistics). The literature in this context is fragmented, relying on distinct algorithms often designed for specific sensors. Most physics-based methods apply thresholds to spectral bands or indices, but their generalizability is limited by the empirical nature of threshold selection. Geometrical and radiative transfer methods demand precise knowledge of acquisition geometry or atmospheric conditions. Multiple-image methods require consistent images from either the same or multiple sensors. Recently, machine learning (ML) and deep learning (DL) approaches have shown potential but require extensive labeled datasets and computational resources. Moreover, most methods target only one surface type, neglecting the impact of the others.

To the best of our knowledge, an approach that jointly copes with the detection of clouds, water, and shadows is still missing. To address this gap, we propose a unified, physics-based approach for joint detection of clouds, water, and shadows in hyperspectral satellite images (APACHE). The proposed method is fully automated and unsupervised, requiring no extensive parameter tuning, labeled data, or multisource images. Rather, it leverages single-image processing and physics-based concepts. APACHE is here proposed within the framework of PRISMA L1 product enhancement, which will be used as a case study, but can be adapted to other sensor data.

The remainder of the article is organized as follows. State-of-the-art analysis of cloud, water, and shadow detection in satellite images is presented in Section II. APACHE methodology is illustrated in Section III. The datasets employed in this work are described in Section IV, whereas Section V presents and discusses experimental results. Conclusion and future directions are provided in Section VI.

II. Previous Works on Cloud, Water, and Shadow Detection

A. Cloud Detection

Cloud detection is crucial for accurately interpreting and properly processing hyperspectral satellite images, as clouds can obscure or considerably alter sensor-reaching radiance from ground surfaces. Thick clouds totally hide surface features, while thin clouds may considerably modify the spectral signatures of the underlying surface materials. In the literature, several different cloud detection algorithms have been developed [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36].

Most physics-based methodologies exploit the fact that clouds typically exhibit high reflectance from VIS through the NIR and SWIR and are characterized by a lower temperature than that of the surface beneath. These methods rely upon thresholding spectral bands or combinations of bands based on spectral indexes, where thresholds are generally empirically selected based on histograms and/or specific sensor characteristics [17], [18], [19], [20], [21], [22], [23], [24], [25], [26]. Among these methods, there are Fmask, specifically developed for Landsat images [17], [18], cloud cover assessment algorithm (ACCA) [19], [20], and cloud detection algorithm-generating method (CDAG) [21]. Within this framework, the cloud detection in hyperspectral images with the atmospheric column water vapor (CHIWAWA) method (see [25]) exploits the atmospheric column water vapor (WV) map, considering that WV is generally underestimated for pixels covered by not too low clouds. Whereas these methods generally work well, their applicability to other sensor data may suffer from the empirical nature of threshold values and due to the lack of specific bands (e.g., thermal band or bands for estimating WV) and generally require parameter tuning and adjustment. Also, some of these methods are prone to misclassify clouds as other bright surfaces such as snow or strong reflectors, especially in urban areas [26].

Radiative transfer-based approaches estimate cloud optical thickness (COT) using lookup tables (LUTs) of ToA radiances for a variety of atmospheric conditions and acquisition geometries, built with radiative transfer codes. Cloud masks are, thus, generated by grouping pixels with nonzero COT values [27], [28]. These methods, however, demand highly accurate LUT construction and knowledge of atmospheric conditions at acquisition time.

More recently, ML methods have been invoked for cloud detection, accounting for cloud spectral (and, in some cases, also textural/spatial) features, such as Markov random field (MRF) framework [29], support vector machine (SVM) [30], neural networks [31], and DL techniques [32], [33]. Whereas, undoubtedly, they provide high detection accuracy, these methods require a very large dataset of both cloud and cloud-free labeled images for training and involve a high computational burden.

Other approaches make use of time series of images acquired from the same sensor or combine images acquired by multiple sensors and seek either for changes over time or with respect to a cloud-free reference image [34], [35], [36]. While benefiting from the information coming from multiple images, these methods are not always applicable because the cloud-free reference images may be characterized by too different acquisition geometries or may be affected by seasonal changes.

B. Water Detection

In the literature, several methods have been applied to detect water bodies in remotely sensed images, both to deliver such pixels to further analyses or to mask them out before a given processing task [37], [38], [39], [40], [41], [42], [43], [44], [45], [46], [47], [48], [49], [50].

Although supervised classification methods may certainly represent a possibility to detect water pixels, they require extensive sampling for each class present in the scene [37]. Thus, water detection has generally been approached by means of ad hoc methods.

The most common approach to water detection is physics-based and exploits the strong absorption performed by water in the NIR and SWIR ranges. Within this framework, simpler

methods just rely upon thresholding bands (generally in the NIR and SWIR), whereas others apply thresholds to combinations of bands based on spectral indexes, as summarized in [37]. Specifically, a variety of indices have been proposed, also driven by the availability of specific spectral bands in various multispectral sensors. Among these, the normalized difference water index (NDWI) [38] and its modified versions, i.e., the green-NDWI (g-NDWI) [39], the modified NDWI (MNDWI) [40], [41], the Sentinel multiband water index (SMBWI) [42], and the optimized spectral threshold water index (OSTWI) [43] have been found to be effective at enhancing water surfaces against others. More sophisticated approaches combine these indices with other spectral analysis techniques (e.g., clustering [44] and rule-based methods [45]). Threshold selection has generally been performed empirically, whereas, more recently, the automated Otsu's method [46], widely employed in computer vision, has been adopted. Whereas undoubtedly effective, thresholding methods often require a data-driven tuning, may struggle in complex scenarios, and often confuse water with dark surfaces such as shadows and urban materials.

Similar to cloud detection, water detection has been recently tackled with ML [47], [48] and DL [49], where texture information has sometimes also been included in the analysis [50]. Similar considerations as above can be made, with ML/DL methods entailing high demands of labeled data and execution times. An in-depth overview of such techniques can be found in [37].

C. Shadow Detection

Shade areas are mostly due to cloud-induced shadows, topographical effects, and buildings in urban areas. In general, shadows obscure surface spectral features and are often confounded with water. Shadow spectral variability is particularly significant because the shadow spectral signature depends on that of the underlying materials.

In the literature, several approaches to shadow detection have been developed, both for masking shadows out and for applying some deshadowing algorithm [26], [51], [52], [53], [54], [55], [56], [57], [58], [59].

Geometric methods utilize sensor position, solar angles, cloud geometry, and terrain configuration to calculate shadow positions at ground [51], [52], [53]. For instance, ray-casting methods determine shadows by projecting clouds on the ground based on cloud height and solar angles. In general, these methods can be effective but rely on detailed and accurate input data—such as a digital elevation model (DEM) and cloud height—which are often difficult to obtain or not readily available.

A few spectral indexes have also been developed for detecting shadows, where shadow low reflectance, especially in VIS and NIR, is exploited for index design, such as the object-based shadow index (OSI) [54] and the cloud shadow index (CSI) [26]. However, the high spectral variability of shadows and the similarity with other dark surfaces (e.g., water) generally require empirical data-driven adjustment of thresholds and parameters or spatial analysis (e.g., to match candidate shadows to detected clouds [26]).

When the satellite sensor has hyperspectral capabilities, other spectral approaches may benefit shadow detection, such as specific coordinate transformation [55] or spectral unmixing [56], [57] where abundance of shade endmembers may be extracted although the results are highly dependent on the endmember choice. Matched filter (MF)-based approaches are also worth mentioning [58], [59], where an MF to the shadow zero-reflectance target spectrum is constructed for shadow detection.

As already mentioned earlier for cloud and water detection, ML and DL methods may also be applied for detecting shadows [29], also incorporating texture information [50]. The limitations are the same as mentioned earlier.

D. Considerations

The importance of detecting clouds, water, and shadow is clearly evident from the various methods proposed in the literature. To the best of our knowledge, a unified approach accounting for the detection of all three kinds of surfaces is still lacking. This aspect is particularly relevant considering the spectral similarity of water and shadows in a great part of the spectrum, as well as the fact that shadows are often detected in relation to clouds.

In this work, a unified physics-based approach for detecting clouds, water, and shadows in hyperspectral images is proposed where detection of the three kinds of surfaces is approached in an automated and fully unsupervised fashion. The method does not require labeled data, and no time-series or multitemporal images are required. Rather, physics-based concepts are combined with an effective single-image processing methodology for sequentially extracting the surfaces of interest. No empirical selection of thresholds or hyperparameter tuning is required.

III. Unified Approach to Detect Cloud, Water, and Shadows for PRISMA Product Enhancement

The general block diagram of APACHE methodology is illustrated in Fig. 1. The workflow consists of several steps, and the aim is to sequentially extract the surfaces of interest at the various steps. Inputs to the various steps are images generated by evaluating the magnitudes of the pixel spectral vectors across different complementary spectral ranges, which are specifically selected based on the characteristic spectral properties of each surface type to be extracted.

For instance, clouds, being highly reflective, exhibit strong and relatively flat spectral signatures across the entire spectral range, making the VIS a suitable range for distinguishing them from other surfaces, characterized by lower and more variable spectral signatures. In contrast, both shadows and water appear dark in the NIR range, which is the proper range to extract them both. Further separating them can be challenging, but we found that including both NIR and SWIR greatly helps their discrimination [26], [43], [60].

The first step, leveraging the VIS range, performs a rough cloud mask extraction, where the thickest cloud pixels are extracted. In the second step, still in the VIS, cloud mask refinement is addressed, with thinner and boundary cloud

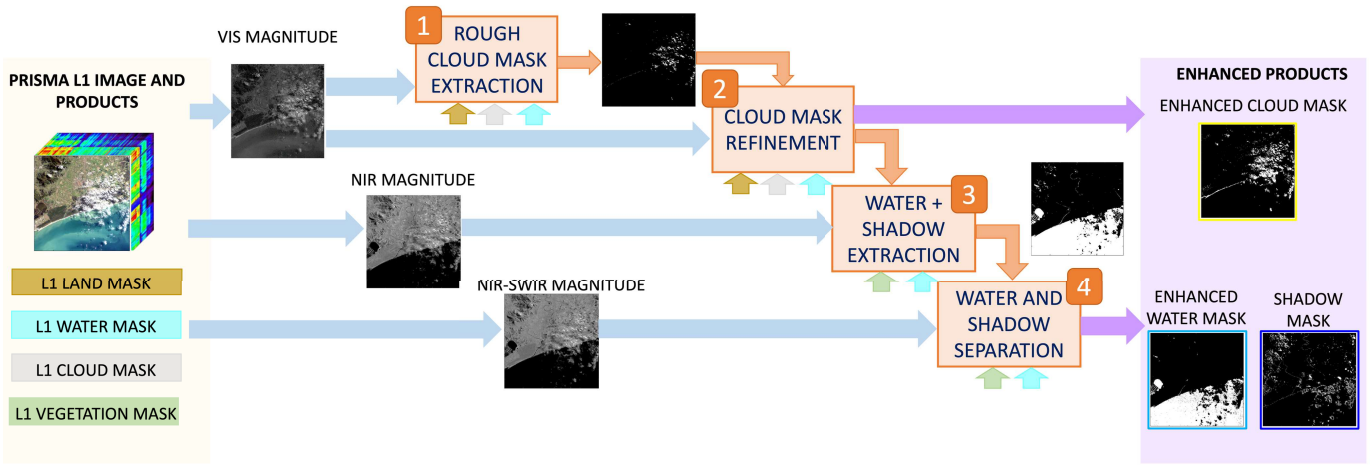


Fig. 1. Block diagram of APACHE sequential methodology proposed for detecting clouds, water, and shadows for PRISMA L1 mask refinement.

pixels being extracted. The third step focuses on the NIR range and aims at extracting both water and shadows. The final step, utilizing both the NIR and SWIR ranges, is aimed at discriminating between water and shadows.

In contrast to many traditional cloud and water detection approaches, which rely on histogram thresholding with empirically derived or data-dependent thresholds, here, we decided to employ, at each step, a nonparametric clustering method originally developed for computer vision applications [61]. This method offers an automated, data-driven solution, avoiding the need for manual threshold selection or tuning. It is important to note that, at each step, we are not concerned with the entire clustering result—specifically, in determining the optimal number of clusters for the given scene and accurately assigning all pixels to these clusters. Instead, our primary interest lies in ensuring that one of the clusters consists exclusively of all pixels belonging to the surface that is to be extracted at that specific step.

A brief overview of the selected clustering method is provided in Section III-A, while Section III-B offers a detailed explanation of each step of the proposed APACHE methodology.

A. Clustering by Search of Density Peaks

This method, originally developed for computer vision applications [61], has proven effective at sequentially extracting the surfaces of interest at each step of the proposed methodology. Basically, it is based on density and distance concepts. Specifically, for each pixel of index n, two quantities are computed: 1) ρ_n, which represents the pixel local density and 2) δ_n, which represents its distance from pixels of higher density [61]. Both quantities depend on the distance d_{nj} between pixel n and each generic pixel j.

The local density ρ_n of nth pixel is defined as in [61]

$$\rho_n = \sum_j \chi\left(d_{nj} - d_c\right) \tag{1}$$

where $\chi(x) = 1$ if $x < 0$ and $\chi(x) = 0$ otherwise. In (1), d_c is the cutoff distance, the choice of which has been shown not to significantly affect results [61].

The pixel distance δ_n is evaluated as the minimum distance between the pixel n and any pixel j with higher density as follows [61]:

$$\delta_n = \min_{j:\rho_j > \rho_n}\left(d_{nj}\right) \tag{2}$$

with $\delta_n = \max_j(d_{nj})$ conventionally assigned to the pixel with highest density.

This method assumes that cluster centers are surrounded by neighbors with lower density and exhibit a relatively large distance from any other pixel with higher density [61]. Indeed, only for pixels that are local or global maxima in terms of the density, δ_n has been observed to be significantly larger than typical nearest-neighbor distances [61]. Cluster centers are, thus, easily identified as those pixels with an anomalously large δ_n. After identification of the cluster centers, the procedure [61] assigns each remaining pixel to the same cluster as its nearest neighbor of higher density, completing the clustering procedure in a single step.

B. APACHE Step-by-Step Description

Let $\mathbf{X} \in \mathbb{R}^{N_b \times N_s \times N_l}$ be the generic hyperspectral data cube with N_b bands, N_s samples and N_l lines. The image spectral magnitude $\mathbf{I} \in \mathbb{R}^{N_s \times N_l}$ evaluated within the spectral range comprised between bands b_1 and b_2 can be defined as follows:

$$I(s,l) = \sqrt{\sum_{b=b_1}^{b_2}\left[\mathbf{X}(b,s,l)\right]^2} \tag{3}$$

where $s = 1,\ldots,N_s$, $l = 1,\ldots,N_l$ and $1 \le b_1 \le b_2 \le N_b$. In our specific focus of PRISMA product enhancement, $\mathbf{X}$ represents the PRISMA L1 calibrated ToA radiance cube.

The following subsections describe, in detail, each step of the proposed APACHE methodology, which is illustrated using a PRISMA hyperspectral image acquired on February 2, 2022, over the city of Pisa (Tuscany, Italy), as depicted in Fig. 2.

1) First Step—Rough Cloud Mask Extraction: The first step aims at roughly separating pixels corresponding to thick clouds from all other image pixels. This rough cloud mask is obtained considering the overall image spectral magnitude in the VIS range, specifically choosing b_1 and b_2 such that the

TABLE I
PARAMETERS OF THE FOUR SEQUENTIAL STEPS

Step #	Band	Wavelength Int.	# Cluster	Init. Criteria per Cluster
I	VIS	$400-700$ nm	3	#1: Max δ_n within L1 cloud pixels #2: Max δ_n within L1 water pixels #3: Max δ_n within L1 land pixels
II	VIS	$400-700$ nm	3	#1: Max δ_n within L1 land pixels #2: Max δ_n within L1 water pixels #3: Max δ_n within L1 cloud pixels
III	NIR	$775-945$ nm	3	#1: Max δ_n within L1 veg. pixels #2: Max δ_n within L1 water pixels #3: Peak of δ_n at lowest ρ_n
IV	NIR-SWIR	$850-1750$ nm	2	#1: Max δ_n within L1 water pixels #2: Peak of δ_n at lowest ρ_n



Fig. 2. RGB representation of the PRISMA hyperspectral image acquired over Pisa, Italy, on February 2, 2022, employed as example image to illustrate the methodology.

corresponding center-band wavelengths are $\lambda_1 = 400$ nm and $\lambda_2 = 700$ nm, respectively (see also Table I, first row).

As previously mentioned, at this step, our goal is to ensure that one of the clusters is composed exclusively of thick cloud pixels. Therefore, identifying three clusters in the image is sufficient: one cluster for cloud pixels, another for water and other dark surfaces, and the last for the remaining pixels, primarily land. To facilitate the selection of cluster centroids, we propose an initialization strategy that leverages PRISMA products, specifically L1 land-use masks, given the focus on PRISMA product enhancement. To ensure adaptability to other sensor data, slight initialization adjustments may be required (e.g., in case some L1 products are not available), which are subject to ongoing work.

The first cluster center is initialized as the pixel with the highest δ_n value among the pixels included in the L1 cloud mask. The second cluster center is selected as the pixel with the highest δ_n among those included in the L1 water mask. As to the third cluster center, it is initialized as the pixel with the highest δ_n among those included in the combined PRISMA L1 masks for bare soil, crop and rangeland, forest, not-vegetated land, and urban component.

Once the cluster centers are initialized, the clustering procedure is executed, as described in Section III-A. The variables and parameters for Step I are summarized in Table I, first row.

Fig. 3(a)–(d) illustrates the application of this first step to the image in Fig. 2. Pixel distances from higher-density pixels are evaluated across the entire VIS image magnitude, and the resulting δ_n values are shown in Fig. 3(a) as an image. Surfaces with high reflectance, such as clouds, but also water foam, and bright surfaces such as certain urban areas exhibit higher δ_n values, indicated by yellow and green shades in the colormap, compared to the overall background (light blue).

Fig. 3(b) shows the distance-to-density plot (here, $(\delta_n)^{1/2}$ is plotted to increase readability). Based on the initialization (see Table I), three cluster centers are automatically selected and represented in the figure by color-coded circles: yellow for the cloud cluster center, light gray for the land cluster center, and dark gray for the water cluster center. Results of the single-step cluster assignment, according to the procedure described in Section III-A, are presented in Fig. 3(c) and (d). These figures display the assignment of each pixel both in the distance-to-density plot and in the scene, using the same color code as the cluster centers. Specifically, pixels belonging to the cloud cluster appear in yellow, those in the land cluster are in light gray, and water cluster pixels are in dark gray. The cluster map in Fig. 3(d) shows that although there is no strict separation between land and water pixels, the primary objective of detecting thick clouds is effectively achieved. These cloud pixels will be masked out and excluded from further consideration in Step II.

2) Second Step—Cloud Mask Refinement: The second step refines the preliminary cloud mask by extracting cloud boundaries and identifying thinner cloud pixels not detected in the previous step. First, pixels in the initial rough cloud mask are excluded from the new evaluation of ρ_n and δ_n values. Then, the procedure from Step I is repeated here (see also Table I, second row). Results are shown in Fig. 4(a)–(d), with the removed rough cloud mask pixels displayed as blank areas in Fig. 4(a) and (d). As is evident from Fig. 4(d), the yellow cluster includes pixels at the boundaries of thicker clouds, as well as some thinner cloud formations and other strong reflectors along the coastline.

For completeness, both cloud masks from Steps I and II are shown in Fig. 5(a) and (b). The rough cloud mask after Step I is displayed in Fig. 5(a), whereas the mask after Step II is shown in Fig. 5(b). The final combined cloud mask from both steps is shown in Fig. 5(c). It is worth noting that some noncloud pixels, such as buildings, highly reflective surfaces, sea foam, and whitecaps, may occasionally be included in the final cloud mask. These pixels should be removed before proceeding to subsequent steps and excluded from the cloud mask. This issue can be addressed using the approach outlined

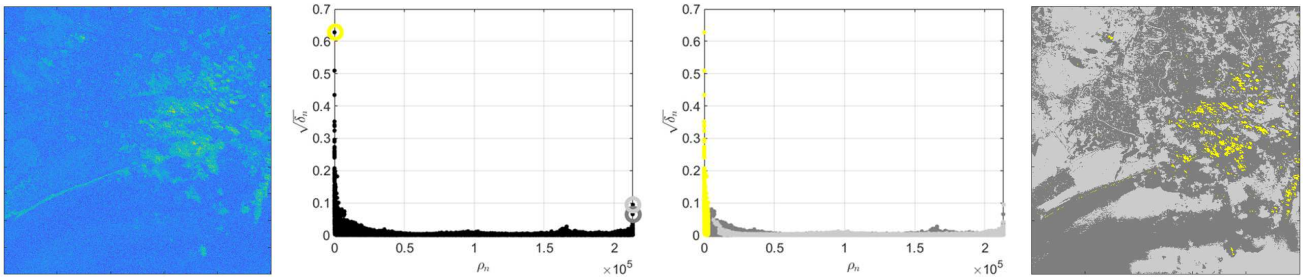


Fig. 3. Step I: extraction of rough cloud mask in the VIS band. (a) Pixel distance ($(\delta_n)^{1/2}$ shown to increase readability) from higher density pixels. (b) Distance-to-density plot (i.e., $(\delta_n)^{1/2}$ plotted against ρ_n), with identified cluster centers highlighted by color-coded circles (yellow for clouds, light gray for land, and dark gray for water). (c) Distance-to-density plot after clustering. (d) Cluster map.

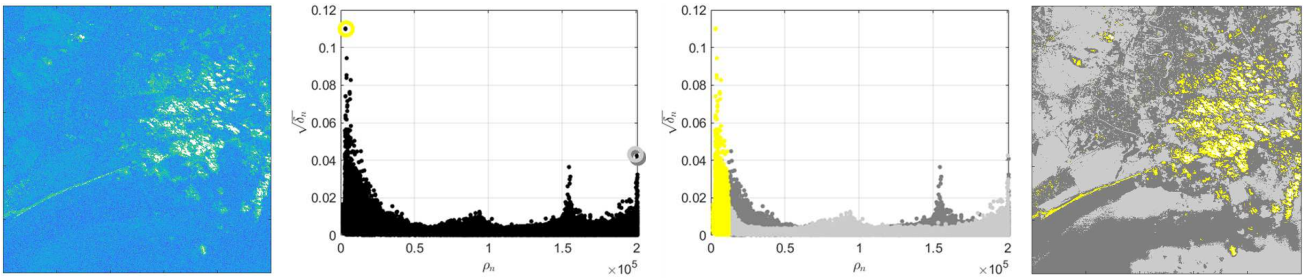


Fig. 4. Step II: refinement of cloud mask in the VIS band. (a) Pixel distance ($(\delta_n)^{1/2}$ shown to increase readability) from higher density pixels–cloud pixels shown as blank pixels. (b) Distance-to-density plot (i.e., $(\delta_n)^{1/2}$ plotted against ρ_n), with identified cluster centers highlighted by color-coded circles (yellow for clouds, light gray for land, and dark gray for water). (c) Distance-to-density plot after clustering. (d) Cluster map–cloud pixels shown as blank pixels.

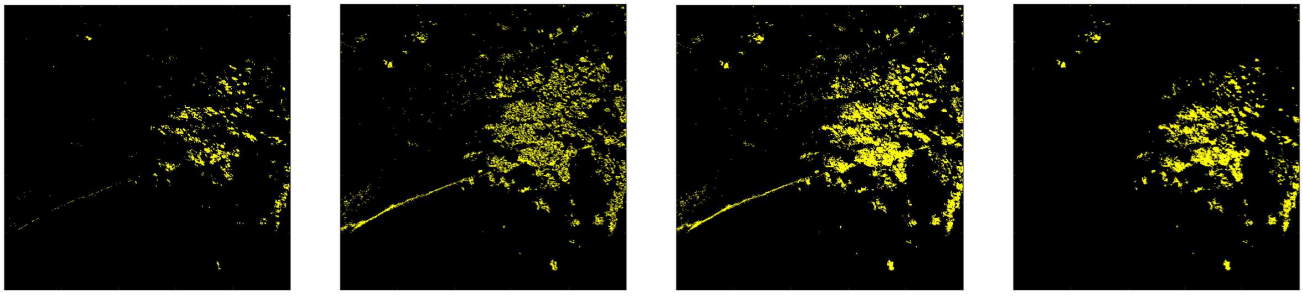


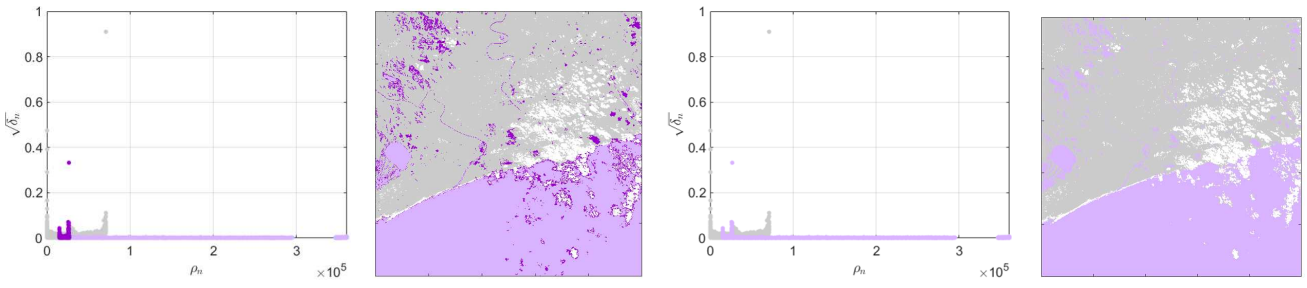
Fig. 5. Extraction of cloud masks (a) rough cloud mask after Step I, (b) refinement of cloud mask after Step II, (c) refined cloud mask after Steps I and II, and (d) polished map after application of [25].

in [25], which leverages the water vapor map. While this approach has not been integrated into our methodology, an example of its application using the water vapor map provided within the PRISMA L2 products is shown in Fig. 5(d).

It is worth noting that when snow and ice are present in the scene, their pixels are likely to be included in the APACHE cloud mask. Although snow detection falls outside the primary scope of this article, we have incorporated an optional module in APACHE to filter snow and ice pixels from the refined cloud mask produced in Step 2. To this end, the normalized difference snow index (NDSI) [62], combined with Otsu's automated thresholding method [46], can be applied specifically to the pixels identified in the refined cloud mask, enabling the identification and removal of snow and ice.

3) Third Step—Dark Surfaces Extraction: The third step aims at generating a water and shadow mask by distinguishing these surfaces from other pixels. The NIR range is selected for this purpose because both water and shadowed areas exhibit low reflectance in this range. Water absorbs radiation in the NIR and thus it is a poor reflector in this range, whereas shadowed regions receive and reflect only diffuse sky radiation, which is lower in the NIR compared to the VIS range.

Before the evaluation of the new ρ_n and δ_n values, all cloud pixels detected in the previous steps are removed. Three cluster centers are then identified in the distance-to-density plot. Due to the strong reflectivity of vegetation in the NIR range, the PRISMA-based initialization selects the first cluster center as the pixel with the highest δ_n value among those in the combined PRISMA L1 masks for forest, crop, and rangeland. The second cluster center is identified as the pixel with the highest δ_n value within the L1 water mask. The third cluster center is initialized as the pixel exhibiting the local maximum of δ_n within the range of lowest ρ_n (see Table I, third row). After having removed clouds and other strong reflectors, we expect this cluster to capture scarcely represented surfaces with



(a) (b) (c) (d)

Fig. 6. Step III: extraction of water and shadow mask in the NIR band (a) distance-to-density plot (i.e., $(\delta_n)^{1/2}$ plotted against ρ_n after clustering (violet and purple for water and shadows, and light gray for the remaining pixels), (b) cluster map–cloud pixels shown as blank pixels, (c) $(\delta_n)^{1/2}$ plotted against ρ_n after merging water plus shadows pixels (violet) and light gray for the remaining pixels, and (d) cluster map-water plus shadow merged cluster (in violet) and light gray for the remaining pixels–cloud pixels shown as blank pixels.

unique reflective properties in the NIR, such as shadows and wetlands.

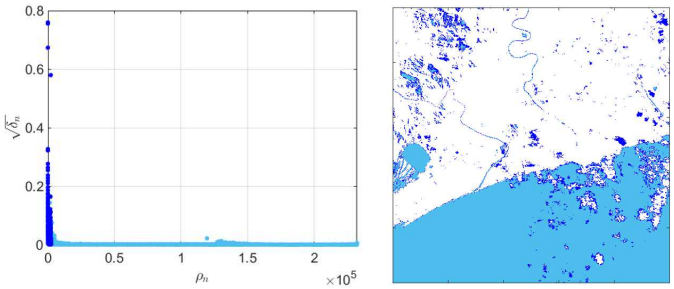
Fig. 6(a) shows the distance-to-density plot after cluster center selection and cluster assignment, whereas Fig. 6(b) presents the corresponding cluster map. In both figures, the first cluster is represented in light gray color and primarily includes vegetation and land areas. The second cluster, shown in violet, mostly covers seawater and the lake on the left, as well as some terrain shadows. The third cluster, depicted in purple, consists of water bodies (mostly rivers) and shadow regions. Cloud pixels, detected at the previous two steps, are left blank, along with highly reflecting pixels. Only the merged second and third clusters, which mostly consist of water and shadow pixels shown in violet in Fig. 6(c) and (d), will be processed in the fourth step to discriminate between water and shadows.

4) Fourth Step—Water/Shadow Discrimination: The goal of Step IV is the final separation between water and shadows, to be applied to all pixels within the merged cluster output from the previous step [see Fig. 6(c) and (d)]. The remaining background pixels, as well as cloud and highly reflective pixels, are discarded. This step focuses on the NIR plus SWIR range (i.e., interval 850–1750 nm), which has been found effective to enhance distinctions between water and shadows in some works [26], [43], [60]. In this step, only two clusters are required. The first cluster center is initialized as the pixel having the highest δ_n among those in the PRISMA L1 water mask. The second cluster center is identified as the local maximum of δ_n with the range of lowest ρ_n values, similar to the approach used in the previous step (see also Table I, fourth row).

Fig. 7(a) shows the distance-to-density plot after cluster center selection and assignment, whereas Fig. 7(b) presents the corresponding cluster map. In both figures, the first cluster, represented in light blue, primarily includes water, while the second cluster, shown in darker blue, mostly captures shadow areas. Cloud pixels along with highly reflecting pixels are left blank, along with background pixels removed in the previous step. While some misassignments remain—such as certain orographic shadows marked as water or cloud shadows over water misidentified as water—the majority of pixels are correctly assigned.

IV. Dataset Description

Since its launch in March 2019, PRISMA has been delivering hyperspectral and panchromatic data with high spectral



(a) (b)

Fig. 7. Step IV: separation of water and shadows in the NIR-SWIR. (a) Distance-to-density plot (i.e., $(\delta_n)^{1/2}$ plotted against ρ_n) after clustering (light blue for water and darker blue for shadows). (b) Cluster map–cloud pixels shown as blank pixels.

TABLE II
PRISMA Hyperspectral Sensor Characteristics [11]

Parameter	VNIR	SWIR
Spectral Range	$400 \div 1010$ nm	$920 \div 2500$ nm
Spectral Bands	63	171
Spectral Res.	≤ 12 nm	
Spatial Res.	30 m	
Swath Width	30 km (FOV $= 2.45°$)	
IFOV	48.34 μrad	

resolution across a wide range of wavelengths, acquiring, downlinking, and archiving up to around 200 30×30 km^2 tiles daily, with nearly global coverage [63].

PRISMA payload includes a hyperspectral camera (HYP) with spectral bands from 400 to 2500 nm, covering the VIS, NIR, and SWIR ranges, at a 30-m spatial resolution. Key characteristics of PRISMA hyperspectral cameras are summarized in Table II (where FOV and IFOV stand for field of view and instantaneous FOV). The payload also includes a panchromatic camera (PAN) with a 5-m resolution from 400 to 700 nm.

The ground segment processes data into various products [64]: L1 provides calibrated ToA spectral radiance data (HYP/PAN), including cloud, land-use, and sun-glint masks; Level 2B offers geo-located BoA spectral radiance data (HYP/PAN); Level 2C provides geo-located surface spectral reflectance with aerosol, water vapor, and cloud characterization; and Level 2-D is the geo-coded Level 2C products. Specifically, L1 land-use mask products are obtained by the PRISMA processor using a hierarchical multiple-class



Fig. 8. RGB mosaic of the primary dataset consisting of 12 PRISMA images acquired over the city of Pisa, Italy, spanning a five-year period from 2020 to 2024.

TABLE III
MAIN DATASET DESCRIPTION

Date (DD/MM/YYYY)	L1 Product Name
04/10/2020	PRS_L1_STD_OFFL_20201004102959_20201004103003_0001
10/06/2021	PRS_L1_STD_OFFL_20210610102029_20210610102033_0001
01/08/2021	PRS_L1_STD_OFFL_20210801101735_20210801101739_0001
07/08/2021	PRS_L1_STD_OFFL_20210807102057_20210807102101_0001
27/10/2021	PRS_L1_STD_OFFL_20211027101806_20211027101810_0001
28/01/2022	PRS_L1_STD_OFFL_20220128102225_20220128102230_0001
02/02/2022	PRS_L1_STD_OFFL_20220202101025_20220202101029_0001
30/05/2023	PRS_L1_STD_OFFL_20230530102106_20230530102110_0001
16/06/2023	PRS_L1_STD_OFFL_20230616101431_20230616101435_0001
25/08/2023	PRS_L1_STD_OFFL_20230825102107_20230825102111_0001
28/10/2023	PRS_L1_STD_OFFL_20231028102428_20231028102432_0001
19/04/2024	PRS_L1_STD_OFFL_20240419102428_20240419102432_0001

classifier aimed at classifying the observed surfaces into a predefined number of spectral categories, including clouds and water [14]. The classifier relies on various spectral index combinations and employs empirical thresholds.

A. Main Dataset

The primary dataset used in this study includes 12 PRISMA images collected over the city of Pisa, Italy, spanning a five-year period from 2020 to 2024. The images were acquired across different seasons and under different atmospheric, illumination, and water conditions. An RGB mosaic of these images is shown in Fig. 8. Details of each image are provided in Table III.

The scene covers the city of Pisa, crossed by the Arno River, which flows into the Tyrrhenian Sea. The Tuscany coastline is visible along the lower edge, with rural areas surrounding the urban area and mountainous terrain in the upper left, where orographic shadows are cast. Additional water bodies include the Serchio River, minor channels, wetlands (e.g., salt marshes), and the Massacciuccoli swampy lake in the bottom left, which appears notably darker than other water bodies. The images show variation in cloud cover (e.g., thin clouds, thick clouds, and veils, with cloud-induced shadows thereof) and water conditions; in one image, the river plumes at the Arno's mouth give the nearby sea a muddy color, with similar turbidity evident in the river itself.

This dataset, collected as part of a broader study, focuses on a well-known geographical area where we have conducted (and continue to conduct) surveys and campaigns. This allows for an easier result interpretation and performance assessment. However, we recognize that all images essentially cover the same scene, including similar surface types and material proportions. To demonstrate the generalizability of our methodology, we also included a secondary dataset with images from different geographical areas and scenarios, described in the following section.

B. Secondary Dataset

As a secondary dataset, we selected 12 PRISMA images capturing a diverse range of scenarios, each characterized by distinct surface features—including bright desert sand, dense and thick vegetation, wetlands, salt marshes, saltworks, lagoons, sediment-rich tropical rivers, snow and ice, and urban areas. This selection allows for a comprehensive assessment of the methodology across varied environments and challenging conditions. More specifically, details are provided in Table IV, and the images are shown in Fig. 9.

C. Ground-Truth Dataset

Five images were manually annotated by the authors with the highest possible accuracy to serve as ground-truth masks for quantitative performance evaluation. Reference information was derived from both RGB and CIR representations and further supported by high-resolution imagery of the same areas. Some degree of uncertainty may still be present—particularly along thin cloud boundaries and in shadowed regions over

TABLE IV
SECONDARY DATASET DESCRIPTION

Date (DD/MM/YYYY)	L1 Product Name	Location
24/11/2021	PRS_L1_STD_OFFL_20211124003316_20211124003321_0001	**Papua New Guinea**
02/11/2021	PRS_L1_STD_OFFL_20211102071709_20211102071713_0001	**Somalia**
13/09/2020	PRS_L1_STD_OFFL_20200913143407_20200913143411_0001	Amazon River
28/01/2024	PRS_L1_STD_OFFL_20240128101031_20240128101036_0001	Venetian lagoon
06/08/2022	PRS_L1_STD_OFFL_20220806100154_20220806100158_0001	Gulf of Naples
23/05/2024	PRS_L1_STD_OFFL_20240523101119_20240523101123_0001	Bracciano Lake
14/01/2020	PRS_L1_STD_OFFL_20200114160935_20200114160940_0001	Finger Lakes
17/09/2021	PRS_L1_STD_OFFL_20210917102722_20210917102726_0001	**Garda Lake**
25/04/2021	PRS_L1_STD_OFFL_20210425165446_20210425165450_0001	**Great Lakes**
15/07/2021	PRS_L1_STD_OFFL_20210715165522_20210715165527_0001	Mississippi River
29/04/2024	PRS_L1_STD_OFFL_20240429130614_20240429130618_0001	Iceland
31/05/2025	PRS_L1_STD_OFFL_20250525103806_20250525103810_0001	Provence

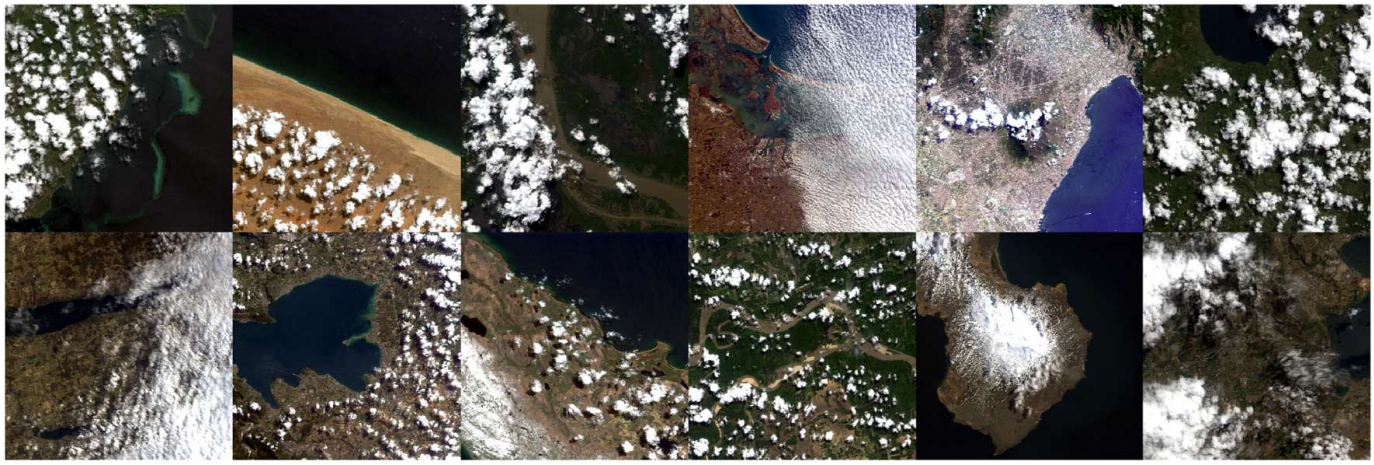


Fig. 9. RGB mosaic of the images from the secondary dataset. First row, from left to right: Papua New Guinea—November 24, 2021; Somalia coast—November 2, 2021; Amazon River—September 13, 2020; Venetian lagoon—January 28, 2024; Naples gulf—August 6, 2022; and Bracciano Lake—May 23, 2024. Second row, from left to right: Finger Lakes—January 14, 2020; Garda Lake—September 17, 2021; Great Lakes—April 25, 2021; Mississippi River—July 15, 2021; Iceland—April 29, 2024; and Provence—May 31, 2025.

dark surfaces (e.g., dense vegetation). These annotated images, selected from both the primary and secondary datasets, are highlighted in bold in Tables III and IV and encompass a range of diverse and complex scenarios.

V. ANALYSIS OF RESULTS

A. Experimental Design

The proposed APACHE methodology, developed in MATLAB (version 9.13.0, R2022b, The MathWorks Inc., Natick, MA, 2022), was applied to images from both datasets, extracting for each image the corresponding cloud, water, and shadow masks. For the local density evaluation, the only free parameter d_c, shown to have minimal impact on the results [61], was set as 10% of the range of $\mathbf{I}$.

First, experimental evidence of the improvements brought by APACHE in enhancing PRISMA L1-based mask products is provided by analyzing the extracted masks for selected portions of both primary and secondary dataset images, focusing on relevant case studies. Specifically, for each selected image portion, the enhanced cloud and water masks, along with the generated shadow mask, are overlaid onto the RGB representation of the corresponding image and visually compared with the original PRISMA L1 mask products. In these visualizations, the detected cloud pixels are highlighted in yellow, water pixels in light blue, and shadow pixels in blue. While this approach does not provide quantitative performance metrics, visual inspection highlights significant improvements in many cases, revealing key insights about the proposed methodology's capabilities.

Second, quantitative performance was assessed using the manually annotated ground-truth dataset. For each surface type, performance was measured in terms of overall accuracy (OA), calculated considering a binary classification framework that distinguishes the given surface (e.g., clouds) from all other pixels (e.g., all the remaining noncloud pixels), as also done in [26]. More specifically, for a given surface of type k (i.e., cloud/water/shadow and snow), OA is defined as

$$OA^{(k)} = \frac{n^{(k)} + n^{(\bar{k})}}{N_s \cdot N_l} \tag{4}$$

where $n^{(k)}$ represents the number of pixels belonging to surface type k, which are correctly classified as such, $n^{(\bar{k})}$ denotes the number of pixels not belonging to surface type k, which are correctly classified as not belonging to it, and $N_s \cdot N_l$ is the total number of image pixels. Note that the numerator corresponds to the total number of correctly classified pixels in the binary classification task for surface type k.

Although a comprehensive comparison with dedicated methods specifically designed for the detection of clouds, water, and shadows lies beyond the scope of this

article—which primarily aims to assess APACHE as a robust, reliable, and automated enhancement tool for PRISMA L1 products—we, nevertheless, include a selection of results from state-of-the-art methods. This provides a quantitative indication of how APACHE performs relative to already benchmarked approaches.

More specifically, cloud and shadow detection is addressed using the cloud and shadow indices proposed by Zhai et al. [26], method hereafter referred to as "HZ," based on the initials of the first author. This method has been chosen because it has shown very promising results for the joint detection of clouds and shadows. Specifically, the HZ cloud index was postprocessed following the procedure described in [26], while the HZ shadow index was used without orographic shadow removal since such shadows are of interest in our analysis. The HZ method requires manual tuning of several thresholds and parameters. In [26], specific parameter values were recommended for a variety of sensors, including Hyperion; given the similar spectral characteristics between Hyperion and PRISMA, we initially adopted those Hyperion default values for all truth images. Subsequently, each parameter was also manually fine-tuned for every truth image, resulting in a nonautomated variant referred to as "HZ tuned." It is clear that we expect to obtain much better results from the finely tuned variant.

For water detection, we employed the well-known NDWI [38], in combination with Otsu's thresholding method [46].

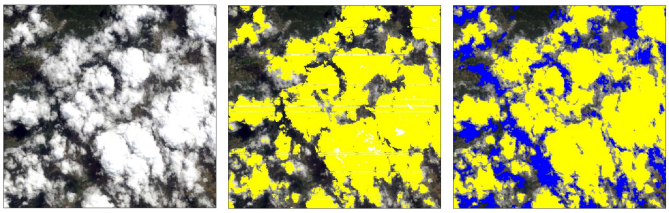
In the truth image containing snow, snow detection was based on the NDSI, complemented by reflectance values in the 0.76–0.9-μm spectral range, as proposed by Hall et al. [62]. For simplicity, this method is referred to as "Hall." The full set of enhanced masks produced by the proposed methodology for all images across all datasets—along with the corresponding L1-based masks and, for the ground-truth dataset, those generated by state-of-the-art methods—is included in the supplementary material. The supplement also provides an example of results obtained with EnMAP data.

B. Results and Discussion—Main Dataset

A selection of results is presented by grouping them into relevant case studies.

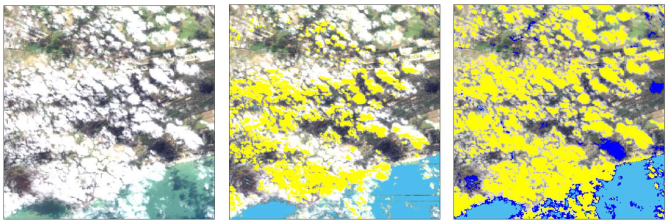
1) Detection of Clouds and Cloud-Induced Shadows: In this section, we analyze selected image portions to assess the capability of the proposed APACHE methodology in detecting clouds of various sizes, distributions, and thicknesses, as well as their associated shadows.

The case study represented in Fig. 10(a) illustrates dense, thick clouds over a dark vegetated area, where the corresponding cloud-induced shadows are visible. PRISMA L1-based cloud mask is shown in Fig. 10(b), while Fig. 10(c) presents both the cloud and shadow masks produced by the proposed methodology. Whereas the two cloud masks are largely similar, the one obtained through APACHE more accurately delineates the full cloud structure and boundaries although some thinner and smaller cloud veils remain undetected. The shadow mask effectively detects cloud-induced shadows, including pixels in between clouds.



(a) (b) (c)

Fig. 10. Portion of the PRISMA image acquired on August 7, 2021. (a) RGB representation. (b) RGB representation with superimposed L1-based cloud mask (in yellow). (c) RGB representation with the superimposed enhanced cloud mask (in yellow) and the produced shadow mask (in blue).



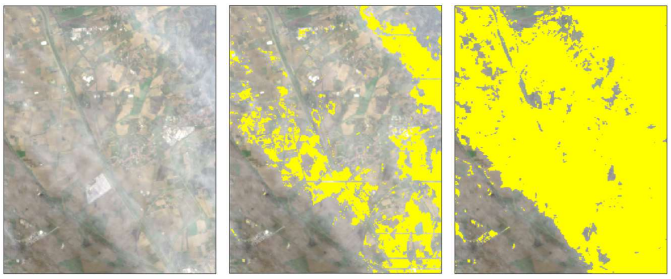
(a) (b) (c)

Fig. 11. Portion of the PRISMA image acquired on February 2, 2022. (a) RGB representation. (b) RGB representation with superimposed L1-based cloud mask (in yellow) and water mask (in light blue). (c) RGB representation with the superimposed enhanced cloud mask (in yellow) and water mask (in light blue), and the produced shadow mask (in blue).

The image portion shown in Fig. 11(a) shows a "mackerel sky" pattern, with small clouds forming densely packed ripples. Significant differences emerge between the L1-based cloud mask [see Fig. 11(b)] and the one obtained using our methodology [see Fig. 11(c)]. While the L1 mask fails to detect many clouds—including some larger ones—the proposed methodology successfully identifies most smaller and thinner clouds, regardless of the underlying surface. However, the shadow mask only partially captures cloud-induced shadows, missing some due to the underlying dark surfaces. Large, shadowed areas cast over the sea remain undetected though some boundary pixels between clouds and water are labeled as shadows—a reasonable outcome considering their darker appearance, though boundary effects may also play a role. For completeness, Fig. 11(b) and (c) also shows the corresponding water masks, both of which include the shaded areas over the sea.

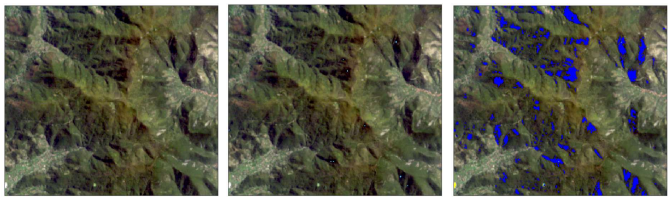
Fig. 12(a) presents an image portion dominated by a thin cloud veil covering most of the scene. Due to variations in thickness, transparency differs across the veil. While the L1-based mask detects only the thickest portions [see Fig. 12(b)], the proposed APACHE methodology successfully identifies most of the cloud veil [see Fig. 12(c)]. No significant cloud-induced shadows are observed in this case.

2) Detection of Terrain Shadows: The previous cases focused on cloud-induced shadows. Here, we analyze a scenario where shadows arise from terrain slopes. Specifically, Fig. 13(a) shows an image portion featuring prominent orographic shadows over Monte Faeta. Since L1-based masks do not explicitly include a shadow mask, they are of limited



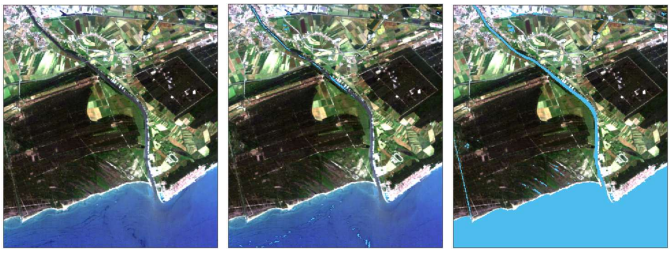
(a) (b) (c)

Fig. 12. Portion of the PRISMA image acquired on October 28, 2023. (a) RGB representation. (b) RGB representation with superimposed L1-based cloud mask (in yellow). (c) RGB representation with the superimposed enhanced cloud mask (in yellow).



(a) (b) (c)

Fig. 13. Portion of the PRISMA image acquired on October 27, 2021. (a) RGB representation. (b) RGB representation with superimposed L1-based water mask (in light blue). (c) RGB representation with the superimposed enhanced water mask (in light blue) and the produced shadow mask (in blue).



(a) (b) (c)

Fig. 14. Portion of the PRISMA image acquired on June 10, 2021. (a) RGB representation. (b) RGB representation with superimposed L1-based water mask (in light blue). (c) RGB representation with the superimposed enhanced water mask (in light blue).

interest in this context; however, Fig. 13(b) displays the L1-based water mask, revealing that only a negligible number of shadow pixels were misclassified as water. The shadow mask produced by APACHE [see Fig. 13(c)] successfully captures most major orographic shadows, demonstrating its potential in nonflat topographic conditions.

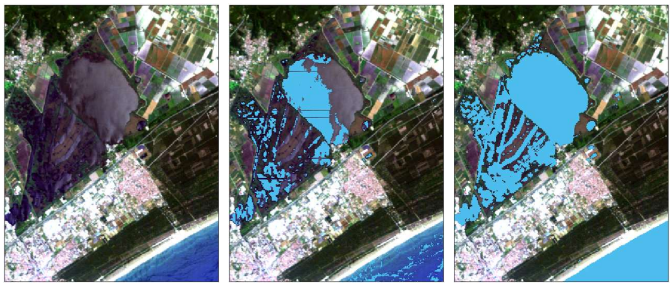
3) Detection of Sea and Inland Water Bodies: This section highlights the potential of the proposed methodology in detecting various water bodies.

Fig. 14(a) depicts the Arno River mouth, where it flows into the Northern Tyrrhenian Sea. While L1-based water masks are typically accurate in this region, in this particular case [see Fig. 14(b)], the mask fails to detect large portions of the river and sea. Conversely, the enhanced water mask [see Fig. 14(c)] achieves a highly precise delineation of not only the Arno River and Tyrrhenian Sea but also smaller water



(a) (b) (c)

Fig. 15. Portion of the PRISMA image acquired on June 10, 2021. (a) RGB representation. (b) RGB representation with superimposed L1-based water mask (in light blue). (c) RGB representation with the superimposed enhanced water mask (in light blue).



(a) (b) (c)

Fig. 16. Portion of the PRISMA image acquired on June 10, 2021. (a) RGB representation. (b) RGB representation with superimposed L1-based water mask (in light blue). (c) RGB representation with the superimposed enhanced water mask (in light blue).

bodies—including the Fiume Morto channel (left-hand side of the scene), ponds, marshes near the coast, and the Navicelli shipyard channel (crossing horizontally the upper part of the scene).

Moving north within the same image, Fig. 15(a) focuses on the Arno River meandering through the town of Pisa and surrounding rural areas. Again, the enhanced mask [see Fig. 15(c)] provides a significant improvement over the L1-based mask [see Fig. 15(b)], accurately capturing the full river shape. The detected feature in the upper right corner corresponds to a small artificial lake used for freshwater fishing.

Further east in the same image, we examine Massaciuccoli Lake, a swampy water body within Tuscany's largest marshland [see Fig. 16(a)]. Due to its muddy water composition, detecting this lake is often challenging. In this case, the L1-based mask [see Fig. 16(b)] captures most of the lake but misses portions of the main basin and smaller surrounding channels and basins. The enhanced mask provides a more defined lake structure and successfully detects additional small ponds in the vicinity.

The image portions in Figs. 17(a) and 18(a) are particularly interesting because the data were acquired after heavy rainfall. Runoff from intense precipitation introduces high concentrations of suspended matter, typically altering the river spectral properties and inducing distinct river plumes at the mouth. The peculiar hue of the sediment-laden water is clearly visible in Fig. 17(a), while a river plume extends into the sea in Fig. 18(a). This presents a significant challenge for



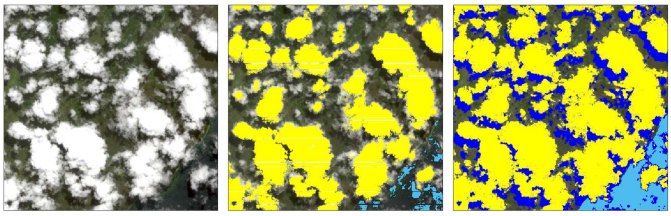
(a) (b) (c)

Fig. 17. Portion of the PRISMA image acquired on June 16, 2023. (a) RGB representation. (b) RGB representation with superimposed L1-based water mask (in light blue). (c) RGB representation with the superimposed enhanced water mask (in light blue).



(a) (b) (c)

Fig. 18. Portion of the PRISMA image acquired on June 16, 2023. (a) RGB representation. (b) RGB representation with superimposed L1-based water mask (in light blue). (c) RGB representation with the superimposed enhanced water mask (in light blue).



(a) (b) (c)

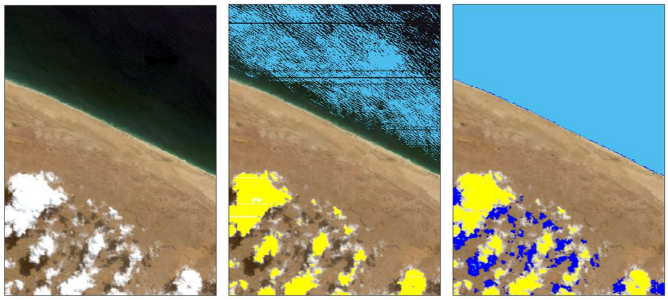
Fig. 19. Portion of the PRISMA image acquired on November 24, 2021. (a) RGB representation. (b) RGB representation with superimposed L1-based cloud mask (in yellow) and water mask (in light blue). (c) RGB representation with the superimposed enhanced cloud mask (in yellow) and water mask (in light blue), and the produced shadow mask (in blue).

conventional water detection algorithms, as suspended sediments considerably modify water spectral properties. Indeed, the L1-based mask fails to detect large portions of the river [see Fig. 17(b)] and entirely misses the sediment-rich plume [see Fig. 18(b)]. The enhanced water mask, however, shows a notable improvement, detecting most of the river course except for its urban segment [see Fig. 17(c)] and accurately capturing the plume-affected sea region [see Fig. 18(c)].

C. Results and Discussion—Secondary Dataset

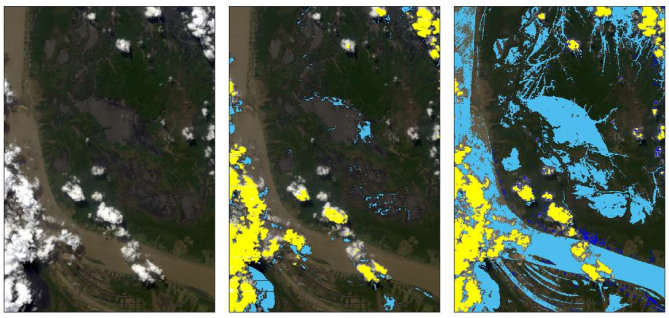
To further evaluate the proposed methodology, results over a secondary dataset are presented, showing performance across diverse scenarios and environments.

1) Detection in Dense, Dark Vegetated Areas: Fig. 19(a) presents an image from the coastal region of Papua New Guinea, where thick clouds and cloud-induced shadows cover dense vegetation and water, resulting in a dark overall appearance. The L1-based cloud mask [see Fig. 19(b)] detects only



(a) (b) (c)

Fig. 20. Portion of the PRISMA image acquired on November 2, 2021. (a) RGB representation. (b) RGB representation with superimposed L1-based cloud mask (in yellow) and water mask (in light blue). (c) RGB representation with the superimposed enhanced cloud mask (in yellow) and water mask (in light blue), and the produced shadow mask (in blue).



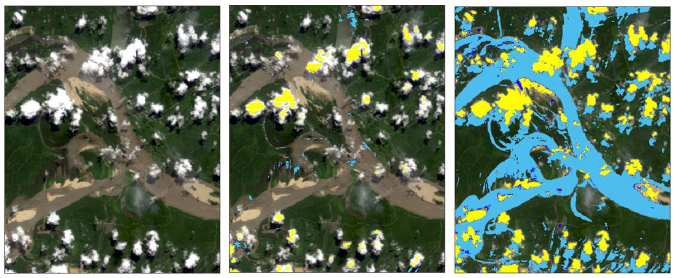
(a) (b) (c)

Fig. 21. Portion of the PRISMA image acquired on September 13, 2020. (a) RGB representation. (b) RGB representation with superimposed L1-based cloud mask (in yellow) and water mask (in light blue). (c) RGB representation with the superimposed enhanced cloud mask (in yellow) and water mask (in light blue), and the produced shadow mask (in blue).

the thickest clouds, missing thinner formations. The L1-based water mask fails to capture much of the sea region in the bottom right corner. The enhanced masks [see Fig. 19(c)] show considerable improvement, precisely delineating clouds, water surfaces, and shadows although shadow detection over water remains less accurate.

2) Detection in Bright, Scarcely Vegetated Landscapes: The image in Fig. 20(a) captures mixed clouds over a bright sandy beach in Somalia. Cloud structure ranges from large and thick to small and thin, and both the L1-based and enhanced cloud masks [see Fig. 20(b) and (c)] perform comparably well. However, the enhanced water mask demonstrates clear improvements, and the shadow mask successfully detects most cloud-induced shadows on the beach though a small fraction of shadow pixels are misclassified as water.

3) Detection in Complex Inland Water Environments: Fig. 21(a) focuses on the Amazon River, with its many nearby lakes, basins, and tributaries, surrounded by the dense, thick Amazon forest. The "whitewater" nature of the Amazon river, characterized by the typical brownish color due to high sediment concentration, makes its detection challenging by empirical thresholding methods like those behind L1-based masks, as is evident in Fig. 21(b). The proposed methodology, which adapts to the scene rather than relying on fixed



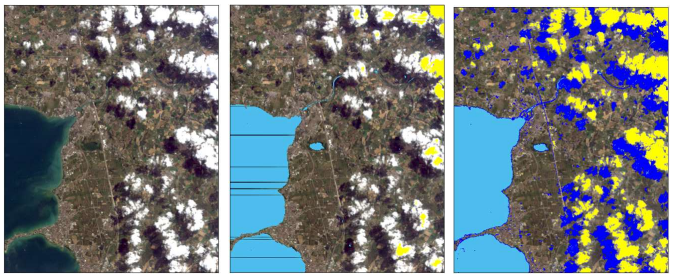
(a) (b) (c)

Fig. 22. Portion of the PRISMA image acquired on July 15, 2021. (a) RGB representation. (b) RGB representation with superimposed L1-based cloud mask (in yellow) and water mask (in light blue). (c) RGB representation with the superimposed enhanced cloud mask (in yellow) and water mask (in light blue), and the produced shadow mask (in blue).



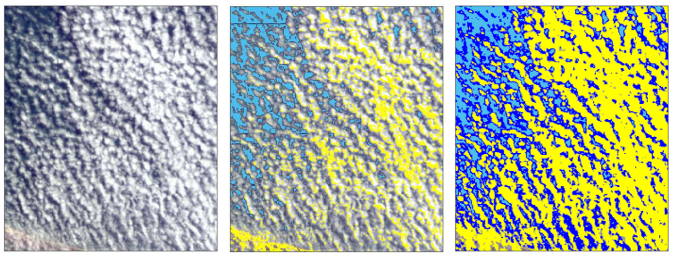
(a) (b) (c)

Fig. 23. Portion of the PRISMA image acquired on May 31, 2025. (a) RGB representation. (b) RGB representation with superimposed L1-based water mask (in light blue). (c) RGB representation with the superimposed enhanced water mask (in light blue).



(a) (b) (c)

Fig. 24. Portion of the PRISMA image acquired on September 19, 2021. (a) RGB representation. (b) RGB representation with superimposed L1-based cloud mask (in yellow) and water mask (in light blue). (c) RGB representation with the superimposed enhanced cloud mask (in yellow) and water mask (in light blue), and the produced shadow mask (in blue).



(a) (b) (c)

Fig. 25. Portion of the PRISMA image acquired on January 28, 2024. (a) RGB representation. (b) RGB representation with superimposed L1-based cloud mask (in yellow) and water mask (in light blue). (c) RGB representation with the superimposed enhanced cloud mask (in yellow) and water mask (in light blue), and the produced shadow mask (in blue).

thresholds, produces a significantly more accurate water mask [see Fig. 21(c)]—some parts of the river and minor basins may indeed not have been completely detected, and some shadow areas have been labeled as water, but it is worth noting that the networks of smaller tributaries and channels do distinctly appear in the enhanced mask. As to cloud detection, the cloud mask obtained with APACHE [see Fig. 21(c)], though missing some smaller and thinner clouds, provides considerable enhancements with respect to the L1-based one [see Fig. 21(b)]. Worse results are provided by the shadow mask, which is tricked by the dark vegetation and water, missing most of the shadows while detecting nonshadow pixels in dark and wetland areas.

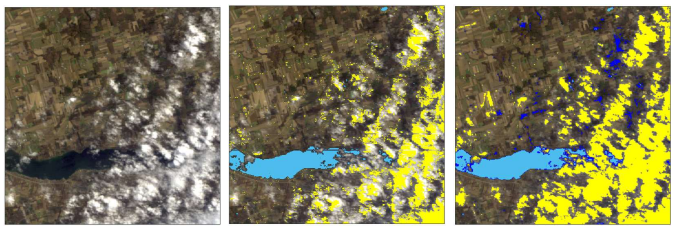
A similar scenario is illustrated in Fig. 22, where the Mississippi River and several of its tributaries flowthrough areas of dense, dark vegetation. As with the Amazon River case, the L1-based cloud and water masks demonstrate low accuracy. In contrast, the APACHE-derived cloud and water masks effectively capture the main cloud formations and water bodies. As in the previous case, shadow detection remains the most challenging aspect, with many shadowed regions mistakenly identified as water.

A particularly distinctive scenario is depicted in Fig. 23, which shows the Berre Lagoon in Provence, France—a brackish body of water influenced by both freshwater and seawater, and heavily impacted by industrial and urban activities, including hydroelectric power generation and salt production. The image specifically highlights the salins, characterized by their distinctive reddish hue caused by carotenoids produced by halophilic microorganisms thriving in high-salinity conditions. The water mask derived from the L1 products fails to accurately identify both the salins and the main lagoon waters. In contrast, the enhanced APACHE water mask successfully captures the extent of both the lagoon and the saline ponds with notable accuracy.

Finally, Fig. 24 illustrates the southern portion of Garda Lake, Italy, embracing the famous Sirmione Castle. Although not especially challenging, this scene features a mix of small urban settlements, agricultural fields, and numerous small clouds scattered across the image, casting shadows on the underlying surface. While the L1-derived water mask successfully captures most of the lake, the corresponding L1 cloud mask fails to detect many of the smaller cloud formations. APACHE significantly improves both the cloud and water masks with high accuracy and also delivers a reliable shadow mask.

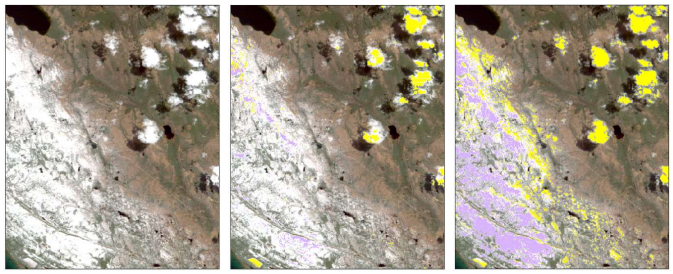
4) Detection of Complex Clouds Formation Over Heterogeneous Surfaces: Fig. 25(a) presents mackerel sky clouds over the Venetian lagoon. A land patch is visible in the bottom left corner, while most clouds cast shadows over the water. The enhanced masks [see Fig. 25(c)] exhibit significant improvements in detecting clouds, water, and shadows compared to the L1-based masks [see Fig. 25(b)].

A similar mackerel sky cloud pattern appears in Fig. 26, which shows the eastern branch of Keuka Lake in the Finger Lakes region, New York, USA. While both the L1



(a) (b) (c)

Fig. 26. Portion of the PRISMA image acquired on January 14, 2020. (a) RGB representation. (b) RGB representation with superimposed L1-based cloud mask (in yellow) and water mask (in light blue). (c) RGB representation with the superimposed enhanced cloud mask (in yellow) and water mask (in light blue), and the produced shadow mask (in blue).



(a) (b) (c)

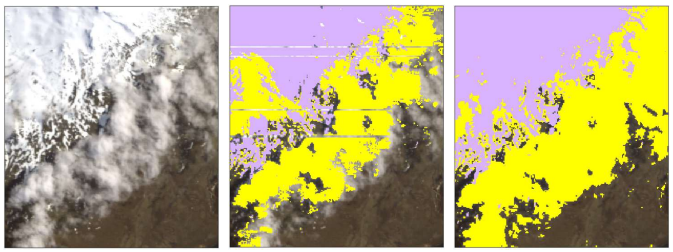
Fig. 27. Portion of the PRISMA image acquired on April 25, 2021. (a) RGB representation. (b) RGB representation with superimposed L1-based cloud mask (in yellow) and snow mask (in violet). (c) RGB representation with the superimposed enhanced cloud mask (in yellow) and snow mask (in violet).

and APACHE water masks yield comparable results—with APACHE offering slight enhancements—the L1-based cloud mask fails to detect the thinner portions of the mackerel cloud pattern. These finer structures are effectively captured by the proposed APACHE methodology. Interpreting the shadow mask is more challenging due to the presence of dark vegetation, which complicates visual assessment. Notably, some thin clouds over the lakes are incorrectly classified as shadows, and several cast shadows appear to have gone undetected.

5) Detection of Clouds in the Presence of Snow and Ice: Although the detection of snow and ice is not an objective of this article, we have included these cases to show that APACHE methodology can be flexibly adapted to handle these complex environments with only minimal adjustments (as described earlier).

The image shown in Fig. 27 is centered on a small peninsula of Lake Superior in the Great Lakes region (Michigan, USA). A snow-covered area is clearly visible in the lower-left portion of the image, just below a small water body (Lake Gratiot), while a group of clouds appears in the upper-right section. The L1-based cloud and snow masks do not perform well, largely failing to detect the corresponding surfaces—particularly the snow mask, which misses most of the snow-covered area. In contrast, APACHE masks exhibit significant improvements in detecting both clouds and snow, with a particularly considerable enhancement in snow detection. Some of the thinner, outer snow regions are mistakenly classified as clouds, but, overall, APACHE provides a far more accurate result.

The image shown in Fig. 28 presents a particularly complex scenario, featuring overlapping cloud and ice formations over the Snafellsjökull glacier in Iceland. The interpretation of both



(a) (b) (c)

Fig. 28. Portion of the PRISMA image acquired on April 29, 2024. (a) RGB representation. (b) RGB representation with superimposed L1-based cloud mask (in yellow) and snow mask (in violet). (c) RGB representation with the superimposed enhanced cloud mask (in yellow) and snow mask (in violet).

the image and the results is inherently challenging due to the presence of uniformly bright surfaces. Nonetheless, some improvements brought by the proposed APACHE methodology are evident. The thinner lower portion of the cloud layer is better captured by APACHE compared to the L1 cloud mask, and the boundaries between ice and clouds in the upper-right section appear more clearly delineated. The APACHE ice mask performs well overall although cloud detection is somewhat confounded near the ice–cloud boundary on the left-hand side of the image—a limitation that is less apparent in the L1 mask. Still, these results highlight APACHE's potential even in highly complex conditions, and they provide a promising foundation for future refinements aimed at even more robust performance in such extreme environments.

D. Quantitative Performance Evaluation and Comparison—Ground-Truth Dataset

This section presents a quantitative performance comparison, expressed in terms of OA (in percentage), for the classification of each target surface against all remaining pixels. The evaluation was conducted across all images in the ground-truth dataset. Results are summarized in Table V, comparing the proposed APACHE method with both the standard L1-based products and the state-of-the-art methods previously described. Corresponding classification masks for all methods and images are provided in the supplementary material, together with truth masks.

Regarding cloud detection, APACHE outperforms the L1-based product in four out of five cases, with particularly significant improvements observed in the Garda Lake and Papua New Guinea images. The only exception is the Somalia image, where APACHE yields slightly lower performance than the L1 product. Compared to the state-of-the-art methods, not only does APACHE consistently outperform the original, automated version of the HZ method across all tested cases, but it also yields better results than the manually fine-tuned HZ variant—except in a single case, the Pisa image, where the tuned HZ achieves higher accuracy.

Regarding water detection, the proposed APACHE method delivers the best performance across all tested images, significantly improving the L1-based products and outperforming the NDWI-based comparison method. In particular, the NDWI approach performs poorly in certain cases—most notably in the Great Lakes image, where it fails to detect the inland water

TABLE V
PERFORMANCE ANALYSIS IN TERMS OF OVERALL ACCURACY

Surface	Method	GREAT LAKES	PISA	GARDA LAKE	PAPUA N.G.	SOMALIA
CLOUDS	**L1 product**	94.85	99.51	91.53	88.01	**96.47**
	HZ	48.72	71.42	70.13	42.15	15.87
	HZ tuned	84.55	**99.63**	97.35	95.36	93.16
	This work	**95.77**	97.69	**98.29**	**95.54**	94.81
WATER	**L1 product**	98.00	98.01	98.76	55.75	66.11
	NDWI + Otsu	70.90	98.34	94.50	27.67	91.48
	This work	**99.11**	**98.79**	**99.47**	**99.20**	**99.60**
SHADOWS	**HZ**	63.94	98.57	71.78	43.01	52.38
	HZ tuned	96.21	**99.44**	92.67	95.22	93.54
	This work	**97.95**	99.28	**97.26**	**95.32**	**98.95**
SNOW	**L1 product**	88.05				
	Hall	90.04				
	This work	**93.39**				

bodies almost entirely, and in the Papua New Guinea image, where it misclassifies large portions of clouds and shadows as water while overlooking shallow water areas.

Shadow detection is best achieved by APACHE in four out of five test images. Compared to both the original automated implementation of the HZ method and its fine-tuned version, APACHE consistently provides significant improvements. While the margin of improvement over the fine-tuned HZ method is smaller, APACHE still outperforms it in most cases—except for the Pisa image, where the fine-tuned HZ shows slightly better performance. It is also worth reminding that no L1 product is available for shadow detection.

Finally, although not a primary objective of this study, snow detection is also addressed. In the examined Great Lake image, APACHE achieves pretty good accuracy although slightly lower than that obtained for the other surface types. The method primarily misses the thinnest snow layers. Notably, APACHE successfully enhances the L1 snow product and outperforms the state-of-the-art Hall method.

E. Results of Application to Other Sensor Data

Although this article focuses on PRISMA images as a case study, we also provide a glimpse of APACHE applicability to other satellite data, such as EnMAP [7]. To maintain focus, these results are presented in the supplementary material. Only minor initialization adjustments were needed to adapt APACHE to EnMAP imagery.

1) Since a vegetation mask is not directly available from EnMAP L1 products, the normalized difference vegetation index (NDVI) [65] combined with Otsu automated thresholding [46] was used in the third step.
2) EnMAP L1 products include a shadow mask, which was directly used in both the third and fourth steps.

No methodological adjustments were needed. Similar considerations can also be made when considering images from other sensors.

As shown in the supplementary material, the minimal modifications required and the good results obtained demonstrate APACHE's flexibility and adaptability to different hyperspectral sensors.

VI. CONCLUSION AND PERSPECTIVES

This article has presented APACHE, a novel, physics-based methodology for joint detection of clouds, water, and shadows in hyperspectral satellite images, with a specific focus on PRISMA L1 product enhancement. The proposed approach sequentially extracts these surfaces in an automated and unsupervised fashion, leveraging the surface spectral properties and a density-based clustering algorithm, while featuring initialization based on PRISMA L1 data. Unlike conventional thresholding-based approaches, which often require extensive empirical tuning, APACHE operates without parameter tuning, dynamically adapting to different spectral scenes, thus being more robust and highly generalizable.

Our methodology was applied to multiple PRISMA images from two significantly different datasets without requiring any adjustments or parameter modifications, showing its strong generalization capability across varying spectral and environmental conditions.

Experimental results have shown the effectiveness of APACHE in detecting the surfaces of interest and refining PRISMA L1 products. Specifically, the enhanced cloud masks exhibited improvements, especially in cloud boundary delineation and thin cloud detection. The enhanced water masks enabled more accurate extraction of water bodies, even in challenging conditions such as plumes, high concentrations of suspended sediments, tropical rivers, and lagoons. The generated shadow masks successfully captured both cloud-induced and terrain shadows, enriching the information content associated with L1 products. A quantitative performance evaluation was conducted on a manually annotated dataset including images from diverse scenarios. On average, APACHE achieved accuracies of 96.42% for cloud detection, 99.23% for water detection, and 97.55% for shadow detection, providing improvements ranging from 2% to 30% over the best-performing comparison method. These results confirmed the effectiveness of APACHE, not only in enhancing the PRISMA L1 products but also in outperforming other state-of-the-art methods. An example of application to other sensor data (i.e., EnMAP) was also provided in the supplementary material.

APACHE represents a significant advancement in hyperspectral satellite image processing and product enhancement, increasing data exploitation possibilities for both scientific and operational uses. Thanks to the improvements introduced, APACHE has the potential to provide high-quality enhanced masks also in support of upcoming spaceborne hyperspectral missions—such as the surface biology and geology (SBG) mission [66] and the Copernicus hyperspectral imaging mission for the environment (CHIME) [67]. Nevertheless, some challenges remain, paving the way for further investigation. The detection of cloud-induced and terrain shadows, especially as regards their discrimination with respect to water and dark surfaces, could be further improved by incorporating spatial correlation with respect to the detected clouds, integrating DEMs, and exploiting full-range hyperspectral signatures for density-based clustering. One feature that deserves further investigation and enhancement is the optional snow/ice detection within the APACHE cloud mask. While it has demonstrated good performance in certain scenarios—particularly where cloud- and snow-/ice-affected areas are separated—it still requires refinement in more challenging cases, such as when both phenomena affect the same pixels. In addition, a strategy to prescreen data as to the potential absence of clouds and water will be introduced so as to enable a simplified APACHE processing.

Further validation using independent datasets from different hyperspectral sensors, along with exploring alternative initialization approaches, could help assess the general applicability of the methodology and identify additional areas for improvement. This would enable handling rare and particularly challenging scenarios, such as lava fields, black sand beaches, and salty lakes. Future work may also explore the integration of more advanced clustering techniques—such as agglomerative hierarchical clustering, which has been successfully applied to Sentinel-2 imagery [68]—as well as the use of contextual and multitemporal information to further improve detection accuracy and reduce residual misclassifications.

ACKNOWLEDGMENT

This project was carried out using ORIGINAL PRISMA Products—© Italian Space Agency (ASI); the products have been delivered under an ASI License to Use.

REFERENCES

[1] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag. Replaces Newslett.*, vol. 1, no. 2, pp. 6–36, Jun. 2013, doi: 10.1109/MGRS.2013.2244672.

[2] S. Chabrillat et al., "The EnMAP spaceborne imaging spectroscopy mission: Initial scientific results two years after launch," *Remote Sens. Environ.*, vol. 315, Dec. 2024, Art. no. 114379, doi: 10.1016/j.rse.2024.114379.

[3] *Compact High Resolution Imaging Spectrometer (CHRIS)*. Accessed: Nov. 8, 2024. [Online]. Available: https://earth.esa.int/eogateway/instruments/chris

[4] *Earth Observing One (EO-1)—Hyperion*. Accessed: Nov. 8, 2024. [Online]. Available: https://www.usgs.gov/centers/eros/science/usgs-eros-archive-earth-observing-one-eo-1-hyperion

[5] L. Chen et al., "An introduction to the Chinese high-resolution Earth observation system: Gaofen-1 7 civilian satellites," *J. Remote Sens.*, vol. 2022, pp. 1–14, Jan. 2022, doi: 10.34133/2022/9769536.

[6] *PRISMA (Hyperspectral Precursor of the Application Mission)*. Accessed: Nov. 8, 2024. [Online]. Available: https://www.asi.it/en/earth-science/prisma/

[7] *EnMAP (Environmental Mapping and Analysis Program)*. Accessed: Nov. 8, 2024. [Online]. Available: https://www.eoportal.org/satellite-missions/enmap/

[8] *DESIS: DLR Earth Sensing Imaging Spectrometer*. Accessed: Nov. 8, 2024. [Online]. Available: https://www.dlr.de/en/research-and-transfer/projects-and-missions/horizons/desis

[9] *Earth Surface Mineral Dust Source Investigation (EMIT)*. Accessed: Nov. 8, 2024. [Online]. Available: https://earth.jpl.nasa.gov/emit/

[10] L. Ansalone et al., "PRISMA mission: Opportunities and main innovations," in *Proc. 24th Italian Assoc. Aeronaut. Astronaut. Int. Conf.*, 2017, pp. 18–22.

[11] R. Loizzo et al., "Prisma: The Italian hyperspectral mission," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2018, pp. 175–178, doi: 10.1109/IGARSS.2018.8518512.

[12] M. F. Buongiorno, M. Musacchio, M. Silvestri, V. Romaniello, C. Spinetti, and F. Rabuffi, "ASI-PRISMA hyperspectral mission for the analysis of geophysical phenomena," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. IGARSS*, Brussels, Belgium, Jul. 2021, pp. 8099–8102, doi: 10.1109/IGARSS47720.2021.9553103.

[13] G. A. Licciardi and M. L. Battagliere, "Prisma hyperspectral data exploitation: The Italian space agency programmes in support of scientific downstream applications, an effort towards the commercial exploitation," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Pasadena, CA, USA, Jul. 2023, pp. 4682–4685, doi: 10.1109/IGARSS52108.2023.10281950.

[14] (2025). *PRISMA ASI Handbook Repository*. Accessed: Feb. 4, 2025. [Online]. Available: https://prisma.asi.it/missionselect/docs

[15] G. Ballesteros, A. Vandenhoeke, L. Antson, and M. Shimoni, "Refining the georeferencing of prisma products using an optical flow methodology," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Pasadena, CA, USA, Jul. 2023, pp. 1736–1739.

[16] S. Maresca and S. Matteoli, "Enhancement of prisma hyperspectral products for improved data exploitation," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Athens, Greece, Jul. 2024, pp. 2239–2242.

[17] Z. Zhu and C. E. Woodcock, "Object-based cloud and cloud shadow detection in Landsat imagery," *Remote Sens. Environ.*, vol. 118, pp. 83–94, Mar. 2012.

[18] Z. Zhu, S. Wang, and C. E. Woodcock, "Improvement and expansion of the fmask algorithm: Cloud, cloud shadow, and snow detection for Landsats 4–7, 8, and Sentinel 2 images," *Remote Sens. Environ.*, vol. 159, pp. 269–277, Mar. 2015.

[19] R. R. Irish, "Landsat 7 automatic cloud cover assessment," *Proc. SPIE*, vol. 4049, pp. 348–355, Aug. 2000.

[20] R. R. Irish, J. L. Barker, S. N. Goward, and T. Arvidson, "Characterization of the Landsat-7 ETM+ automated cloud-cover assessment (ACCA) algorithm," *Photogramm. Eng. Remote Sens.*, vol. 72, no. 10, pp. 1179–1188, Oct. 2006.

[21] L. Sun et al., "A cloud detection algorithm-generating method for remote sensing data at visible to short-wave infrared wavelengths," *ISPRS J. Photogram. Remote Sens.*, vol. 124, pp. 70–88, Feb. 2016, doi: 10.1016/j.isprsjprs.2016.12.005.

[22] W. B. Rossow and L. C. Garder, "Cloud detection using satellite measurements of infrared and visible radiances for ISCCP," *J. Climate*, vol. 6, no. 12, pp. 2341–2369, Dec. 1993.

[23] L. L. Stowe, P. A. Davis, and E. P. McClain, "Scientific basis and initial evaluation of the CLAVR-1 global clear/cloud classification algorithm for the Advanced Very High Resolution Radiometer," *J. Atmos. Ocean. Technol.*, vol. 16, no. 6, pp. 656–681, Jun. 1999.

[24] K. T. Kriebel, G. Gesell, M. Kaästner, and H. Mannstein, "The cloud analysis tool APOLLO: Improvements and validations," *Int. J. Remote Sens.*, vol. 24, no. 12, pp. 2389–2408, Jan. 2003.

[25] A. Alakian, "Cloud detection in hyperspectral images with atmospheric column water vapor: Application to PRISMA and AVIRIS-NG images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5510824, doi: 10.1109/TGRS.2024.3369401.

[26] H. Zhai, H. Zhang, L. Zhang, and P. Li, "Cloud/shadow detection based on spectral indices for multi/hyperspectral optical remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 144, pp. 235–253, Oct. 2018.

[27] M. D. King et al., "Airborne scanning spectrometer for remote sensing of cloud, aerosol, water vapor, and surface properties," *J. Atmos. Ocean. Technol.*, vol. 13, no. 4, pp. 777–794, Aug. 1996.

[28] L. Guanter, R. Richter, and H. Kaufmann, "On the application of the MODTRAN4 atmospheric radiative transfer code to optical remote sensing," *Int. J. Remote Sens.*, vol. 30, no. 6, pp. 1407–1424, Mar. 2009.

[29] S. Le Hégarat-Mascle and C. André, "Use of Markov random fields for automatic cloud/shadow detection on high resolution optical images," *ISPRS J. Photogramm. Remote Sens.*, vol. 64, no. 4, pp. 351–366, Jul. 2009.

[30] C. Latry, C. Panem, and P. Dejean, "Cloud detection with SVM technique," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jan. 2007, pp. 448–451.

[31] J.-D. Jang, A. A. Viau, F. Anctil, and E. Bartholomé, "Neural network application for cloud detection in SPOT VEGETATION images," *Int. J. Remote Sens.*, vol. 27, no. 4, pp. 719–736, Feb. 2006.

[32] M. Segal-Rozenhaimer, A. Li, K. Das, and V. Chirayath, "Cloud detection algorithm for multi-modal satellite imagery using convolutional neural-networks (CNN)," *Remote Sens. Environ.*, vol. 237, Feb. 2020, Art. no. 111446.

[33] Y. Li, W. Chen, Y. Zhang, C. Tao, R. Xiao, and Y. Tan, "Accurate cloud detection in high-resolution remote sensing imagery by weakly supervised deep learning," *Remote Sens. Environ.*, vol. 250, Dec. 2020, Art. no. 112045.

[34] A. Lyapustin, Y. Wang, and R. Frey, "An automatic cloud mask algorithm based on time series of MODIS measurements," *J. Geophys. Res., Atmos.*, vol. 113, no. D16, pp. 1–15, Aug. 2008.

[35] O. Hagolle, M. Huc, D. V. Pascual, and G. Dedieu, "A multi-temporal method for cloud detection, applied to FORMOSAT-2, VENμS, Landsat and Sentinel-2 images," *Remote Sens. Environ.*, vol. 114, no. 8, pp. 1747–1755, Aug. 2010.

[36] G. Mateo-García, L. Gómez-Chova, J. Amorós-López, J. Muñoz-Marí, and G. Camps-Valls, "Multitemporal cloud masking in the Google Earth engine," *Remote Sens.*, vol. 10, no. 7, p. 1079, Jul. 2018.

[37] T. V. Bijeesh and K. N. Narasimhamurthy, "Surface water detection and delineation using remote sensing images: A review of methods and algorithms," *Sustain. Water Resour. Manage.*, vol. 6, no. 4, pp. 1–23, Aug. 2020, doi: 10.1007/s40899-020-00425-4.

[38] B.-C. Gao, "NDWI—A normalized difference water index for remote sensing of vegetation liquid water from space," *Remote Sens. Environ.*, vol. 58, no. 3, pp. 257–266, Dec. 1996, doi: 10.1016/s0034-4257(96)00067-3.

[39] S. K. McFeeters, "The use of the normalized difference water index (NDWI) in the delineation of open water features," *Int. J. Remote Sens.*, vol. 17, no. 7, pp. 1425–1432, May 1996.

[40] H. Xu, "Modification of normalised difference water index (NDWI) to enhance open water features in remotely sensed imagery," *Int. J. Remote Sens.*, vol. 27, no. 14, pp. 3025–3033, Jul. 2006.

[41] Y. Du, Y. Zhang, F. Ling, Q. Wang, W. Li, and X. Li, "Water bodies' mapping from Sentinel-2 imagery with modified normalized difference water index at 10-m spatial resolution produced by sharpening the SWIR band," *Remote Sens.*, vol. 8, no. 4, p. 354, Apr. 2016, doi: 10.3390/rs8040354.

[42] Z. Su et al., "A new and robust index for water body extraction from Sentinel-2 imagery," *Remote Sens.*, vol. 16, no. 15, p. 2749, Jul. 2024, doi: 10.3390/rs16152749.

[43] C. Li et al., "New automated method for extracting river information using optimized spectral threshold water index," *Arabian J. Geosci.*, vol. 12, no. 1, pp. 1–12, Jan. 2019, doi: 10.1007/s12517-018-4124-z.

[44] T. Halder et al., "A hybrid approach for water body identification from satellite images using NDWI mapping and histogram of gradients," *Innov. Syst. Softw. Eng.*, vol. 20, no. 2, pp. 111–120, Jun. 2024, doi: 10.1007/s11334-021-00414-6.

[45] F. Chen, X. Chen, T. Van de Voorde, D. Roberts, H. Jiang, and W. Xu, "Open water detection in urban environments using high spatial resolution remote sensing imagery," *Remote Sens. Environ.*, vol. 242, Jun. 2020, Art. no. 111706, doi: 10.1016/j.rse.2020.111706.

[46] N. Otsu, "A threshold selection method from gray-level histograms," *IEEE Trans. Syst., Man, Cybern.*, vol. SMC-9, no. 1, pp. 62–66, Jan. 1979, doi: 10.1109/TSMC.1979.4310076.

[47] P. Frazier and K. Page, "Water body detection and delineation with Landsat TM data," *Photogrammetric Eng. Remote Sens.*, vol. 66, no. 12, pp. 1461–1467, 2000.

[48] X. Huang, C. Xie, X. Fang, and L. Zhang, "Combining pixel- and object-based machine learning for identification of water-body types from urban high-resolution remote-sensing imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 5, pp. 2097–2110, May 2015, doi: 10.1109/JSTARS.2015.2420713.

[49] F. Isikdogan, A. C. Bovik, and P. Passalacqua, "Surface water mapping by deep learning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 11, pp. 4909–4918, Nov. 2017, doi: 10.1109/JSTARS.2017.2735443.

[50] C. G. Tate, J. Cave, and R. L. Moyers, "Segmenting water and shadow regions within WorldView imagery using local binary patterns," *J. Appl. Remote Sens.*, vol. 16, no. 3, Sep. 2022, Art. no. 34532, doi: 10.1117/1.jrs.16.034532.

[51] J. J. Simpson and J. R. Stitt, "A procedure for the detection and removal of cloud shadow from AVHRR data over land," *IEEE Trans. Geosci. Remote Sens.*, vol. 36, no. 3, pp. 880–897, May 1998, doi: 10.1109/36.673680.

[52] J. J. Simpson, Z. Jin, and J. R. Stitt, "Cloud shadow detection under arbitrary viewing and illumination conditions," *IEEE Trans. Geosci. Remote Sens.*, vol. 38, no. 2, pp. 972–976, Mar. 2000, doi: 10.1109/36.841979.

[53] Y. Luo, A. P. Trishchenko, and K. V. Khlopenkov, "Developing clear-sky, cloud and cloud shadow mask for producing clear-sky composites at 250-meter spatial resolution for the seven MODIS land bands over Canada and North America," *Remote Sens. Environ.*, vol. 112, no. 12, pp. 4167–4185, Dec. 2008, doi: 10.1016/j.rse.2008.06.010.

[54] H. Fu, T. Zhou, and C. Sun, "Object-based shadow index via illumination intensity from high resolution satellite images over urban areas," *Sensors*, vol. 20, no. 4, p. 1077, Feb. 2020, doi: 10.3390/s20041077.

[55] E. Ashton, B. D. Wemett, R. A. Leathers, and T. V. Downes, "A novel method for illumination suppression in hyperspectral images," *Proc. SPIE*, vol. 6966, pp. 84–91, Apr. 2008.

[56] J. W. Boardman. (1993). *Automating Spectral Unmixing of AVIRIS Data Using Convex Geometry Concepts*. [Online]. Available: https://api.semanticscholar.org/CorpusID:140591692

[57] G. Zhang, D. Cerra, and R. Müller, "Shadow detection and restoration for hyperspectral images based on nonlinear spectral unmixing," *Remote Sens.*, vol. 12, no. 23, p. 3985, Dec. 2020, doi: 10.3390/rs12233985.

[58] R. Richter and A. Müller, "De-shadowing of satellite/airborne imagery," *Int. J. Remote Sens.*, vol. 26, no. 15, pp. 3137–3148, Aug. 2005, doi: 10.1080/01431160500114664.

[59] S. M. Adler-Golden, M. W. Matthew, G. P. Anderson, G. W. Felde, and J. A. Gardner, "Algorithm for de-shadowing spectral imagery," *Proc. SPIE*, vol. 4816, pp. 203–210, Feb. 2002, doi: 10.1117/12.451691.

[60] W. Jiang et al., "An effective water body extraction method with new water index for Sentinel-2 imagery," *Water*, vol. 13, no. 12, p. 1647, Jun. 2021, doi: 10.3390/w13121647.

[61] A. Rodriguez and A. Laio, "Clustering by fast search and find of density peaks," *Science*, vol. 344, no. 6191, pp. 1492–1496, Jun. 2014, doi: 10.1126/science.1242072.

[62] D. K. Hall, G. A. Riggs, and V. V. Salomonson, "Development of methods for mapping global snow cover using Moderate Resolution Imaging Spectroradiometer data," *Remote Sens. Environ.*, vol. 54, no. 2, pp. 127–140, Nov. 1995.

[63] R. Guarini, R. Loizzo, F. Longo, S. Mari, T. Scopa, and G. Varacalli, "Overview of the prisma space and ground segment and its hyperspectral products," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Fort Worth, TX, USA, Jul. 2017, pp. 431–434, doi: 10.1109/IGARSS.2017.8126986.

[64] R. Guarini et al., "Prisma hyperspectral mission products," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2018, pp. 179–182, doi: 10.1109/IGARSS.2018.8517785.

[65] J. W. Rouse, R. H. Haas, J. A. Scheel, and D. W. Deering, "Monitoring vegetation systems in the great plains with ERTS," in *Proc. 3rd Earth Resource Technol. Satell. (ERTS) Symp.*, vol. 1, May 1974, pp. 48–62.

[66] K. Cawse-Nicholson et al., "Surface biology and geology imaging spectrometer: A case study to optimize the mission design using intrinsic dimensionality," *Remote Sens. Environ.*, vol. 290, May 2023, Art. no. 113534, doi: 10.1016/j.rse.2023.113534.

[67] M. Rast, J. Nieke, J. Adams, C. Isola, and F. Gascon, "Copernicus hyperspectral imaging mission for the environment (chime)," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. IGARSS*, Brussels, Belgium, Jul. 2021, pp. 108–111, doi: 10.1109/IGARSS47720.2021.9553319.

[68] M. C. R. Cordeiro, J.-M. Martinez, and S. Peña-Luque, "Automatic water detection from multidimensional hierarchical clustering for Sentinel-2 images and a comparison with level 2A processors," *Remote Sens. Environ.*, vol. 253, Feb. 2021, Art. no. 112209, doi: 10.1016/j.rse.2020.112209.

Open Access funding provided by 'Consiglio Nazionale delle Ricerche' within the CRUI CARE Agreement